

Mohammad Yazdani

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EDUCATION

UNIVERSITY OF TEHRAN

M.Sc., Process Engineering — GPA: 3.65/4

Tehran, Iran

Sep 2023 – Present

Thesis: *Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO₂ Capture Process for Steam Methane Reforming Flue Gas*

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Sc., Chemical Engineering — GPA: 3.63/4

Tehran, Iran

Sep 2019 – Sep 2023

Thesis: *From Model to Market: A Feasibility Study of Acrylic Ester Production*

RESEARCH INTERESTS

- Process Systems Engineering (PSE) • Process Modeling, Simulation & Digital Twins • Mathematical Optimization • Carbon Capture, Utilization & Storage (CCUS) • Techno-Economic & Sustainability Analysis • Artificial Intelligence for Chemical Engineering

PUBLICATIONS

JOURNAL ARTICLE

- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. “Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO₂ Capture Process for Steam Methane Reforming Flue Gas”. **Accepted** at *Journal of CO₂ Utilization*. **(Quartiles: Q1; IF: 8.6)**
- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. “Techno-Economic Assessment of an Optimized Integrated Adsorption-Cryogenic CO₂ Capture Process for Steam-Methane Reforming Flue Gas”. **Under review** at the *International Journal of Greenhouse Gas Control*. **(Quartiles: Q1; IF: 5.9)**

CONFERENCE ARTICLES

- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. “Surrogate-Assisted Multi-Objective Optimization of VPSA for CO₂ Capture from Steam Methane Reforming Flue Gas.” The 13th International Chemical Engineering Congress & Exhibition (ICHEC 2026), Tehran, Iran, 2026.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. “Integrated Waste Heat Recovery and Adsorption System for SO₂ Removal from Flue Gas.” The 13th International Chemical Engineering Congress & Exhibition (ICHEC 2026), Tehran, Iran, 2026.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. “Simulation of a Hybrid Vacuum Swing Adsorption–Cryogenic Process for CO₂ Removal from Flue Gas.” Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- Charoughchi, A., and **Yazdani, M.** “Simulation of Sulfur Dioxide (SO₂) Pollutant Adsorption from High-Sulfur Fuel Combustion.” Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. “Development of Machine Learning-Based Surrogate Models for the Optimization of Direct Air Capture (DAC) Process.” 6th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. “Comparative Study of Response Surface Methodology (RSM) and Random Forest (RF) Models for Predicting SO₂ Recovery from Flue Gas in an Adsorption Process.” 6th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.

RESEARCH EXPERIENCE

UNIVERSITY OF TEHRAN

Research Assistant

Tehran, Iran

Jun 2024 - Present

- Developed automated workflows for process modeling, simulation, optimization, and digital process engineering using Aspen Adsorption, Aspen Plus, Python, and machine learning.
- Implemented mathematical optimization and AI-enabled surrogate modeling for process design and techno-economic assessment.
- Reduced computational time for process optimization by replacing high-fidelity simulations with surrogate models, enabling rapid exploration of hundreds of operating conditions.

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Tehran, Iran

Research Assistant

Mar 2023 - Sep 2023

- Led an independent techno-economic and market analysis of five competing industrial production technologies for acrylate esters (Dow, BASF, Toyo Soda, Distillers, Arkema), synthesizing process data, economic modeling, and patent life-cycle analysis into a unified evaluation framework.
- Designed and applied a multi-criteria decision-making methodology (AHP) to systematically compare licensing technologies across economic, market, and technical criteria, resulting in a rigorously justified technology selection.
- Developed a detailed process simulation in Aspen Plus and completed equipment design for the selected process, demonstrating the ability to translate technical literature into a validated, working process model.

TEACHING EXPERIENCE

UNIVERSITY OF TEHRAN

Tehran, Iran

Teaching Assistant – Advanced Engineering Mathematics

Oct 2024 - Feb 2025

- Assisted in delivering tutorials and mentoring students in advanced engineering mathematics, guiding analytical and numerical solution methods.
- Developed and delivered hands-on MATLAB workshops covering numerical methods, mathematical modeling, and engineering problem solving.

UNIVERSITY OF TEHRAN

Tehran, Iran

Teaching Assistant – Advanced Adsorption Processes

Oct 2025 - Feb 2026

- Assisted in teaching advanced adsorption processes by mentoring students in adsorption theory, process modeling, and simulation concepts.
- Developed and delivered hands-on Aspen Adsorption workshops covering process model development, cycle simulation, and process performance evaluation.

PROFESSIONAL EXPERIENCE

PETRO RAMAN TECHNOLOGY (PRT) CO.

Tehran, Iran

Process Engineer

Jun 2023 - Present

- Led the Process Engineering discipline for feasibility, conceptual design, basic and detailed engineering, and debottlenecking studies across petrochemical projects.
- Performed process simulation, engineering calculations, equipment specification, and engineering document review.
- Collaborated with multidisciplinary engineering teams and licensors throughout project execution.

SHAZAND PETROCHEMICAL COMPANY (AR.P.C)

Shazand, Iran

Internship

Jul 2022 - Sep 2022

- Gained hands-on process engineering experience in an operating Olefin unit, working from Technip licensor design documents to perform equipment sizing, hydraulic calculations, and process simulation (Aspen HYSYS/Plus), with direct exposure to live plant operations.

TECHNICAL SKILLS

Programming: Python, MATLAB, FORTRAN

Process Simulation & Engineering Software: Aspen Plus, Aspen HYSYS, AVEVA Process Simulation, DWSIM, UniSim, Aspen Custom Modeler, Aspen Adsorption, Aspen Process Economic Analyzer, Aspen Flare System Analyzer, Aspen Exchanger Design & Rating, HTRI, PIPEFLO, AFT Impulse & Fathom, CONVAL, COMSOL Multiphysics

General: AutoCAD, Microsoft Office, Adobe Products, Figma

Languages: Persian (native), English (C1 – TOEFL iBT 100: R27, L25, S24, W24)

LICENSES & CERTIFICATIONS

AI FOR EVERYONE, BY ANDREW NG

Nov 2023

- Issued by DeepLearning.AI (Show [credential](#))

AI PYTHON FOR BEGINNERS, BY ANDREW NG

Oct 2024

- Issued by [DeepLearning.AI](#)

MACHINE LEARNING SPECIALIZATION, BY ANDREW NG

Nov 2024 - Feb 2025

- Issued by DeepLearning.AI, Stanford University (Show [credential](#))

REFERENCES

DR. SHOHREH FATEMI

M.Sc. Supervisor

- Professor at Department of Polymer and Chemical Engineering, University of Tehran, Tehran, Iran, shfatemi@ut.ac.ir

DR. MOHAMMAD FAKHROLESLAM

M.Sc. Co-Supervisor

- Assistant Professor at [Tarbiat Modares University](#), Tehran, Iran, fakhroleslam@modares.ac.ir

DR. NOROLLAH KASIRI

B.Sc. Supervisor

- Professor at Iran University of Science and Technology, Tehran, Iran, kasiri@iust.ac.ir